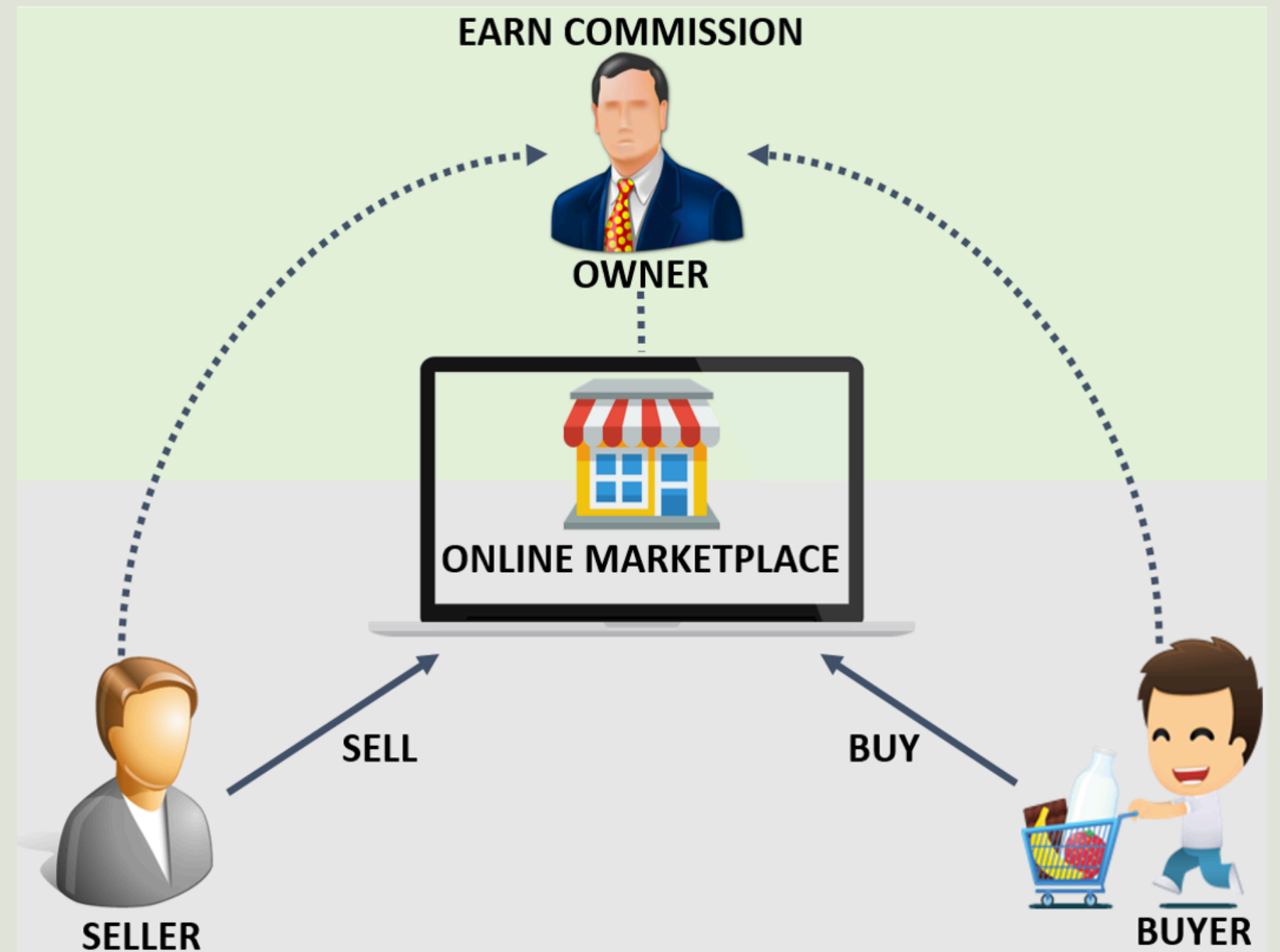


ONLINE MARKET PLACE

Product Recommendation Graph

CRUD



PROBLEM STATEMENT

- **Traditional shopping is time-consuming and location-dependent**
- **Limited product availability in physical stores**
- **Difficulty in comparing prices and options**
- **Need for a scalable, user-friendly online solution**

REAL WORLD APPLICATION



- Used by platforms like Amazon, Flipkart, eBay
- Supports online shopping for electronics, clothing, groceries, etc.
- Helps small businesses reach a global audience
- Enables digital payments and doorstep delivery

DATA STRUCTURES USED

- **Arrays / Lists → Store product collections**
- **Hash Tables → Fast user authentication & lookup**
- **Linked Lists → Manage orders dynamically**
- **Trees (BST/Category Tree) → Product categorization**
- **Graphs (optional) → Recommendation systems**

SYSTEM DESIGN



- **Frontend → User interface (web/mobile app)**
- **Backend → Server handling logic and requests**
- **Database → Stores users, products, orders**
- **APIs → Connect frontend and backend**
- **Payment Gateway → Secure transactions**

ALGORITHM

- **Search Algorithm → Find products quickly**
- **Sorting Algorithm → Price, rating, popularity**
- **Recommendation Algorithm → Suggest products**
- **Authentication Algorithm → Secure login**
- **Order Processing Logic → Checkout workflow**

IMPLEMENTATION



- Languages: Java / Python / JavaScript
- Database: MySQL / MongoDB
- Frameworks: React / Node.js / Django (optional)
- Modules:
- User Registration/Login
- Product Management
- Cart System
- Payment Integration

FEATURES

- **Secure user login and registration for buyers and sellers**
- **Easy product search and filtering options**
- **Shopping cart and secure payment system**
- **Order tracking and history for users**
- **Reviews, ratings, and admin management for better control**



DEMO OUTPUT

```
=====
ONLINE MARKETPLACE GRAPH SYSTEM
=====
1. Add Node
2. Delete Node
3. Update Node
4. Search
5. Display
6. Add Recommendation
7. Exit
```

```
Enter your choice: 1

Enter Product ID: 101
Enter Product Name: phone
→ Product Added Successfully!
```

Enter your choice: 1

Enter Product ID: 110

Enter Product Name: glass

Product Added Successfully!

Enter your choice: 2

Enter Product ID to Delete: 110

Product Deleted Successfully!

Enter your choice: 3

Enter Product ID to Update: 101

Enter New Name: pouchs

Product Updated Successfully!

Enter your choice: 4

Enter Product ID to Search: 101

Found Product ID: 101 | Name: pouchs

```
Enter your choice: 5
```

```
===== PRODUCT LIST =====
```

```
ID: 101 | Name: pouchs
```

```
Recommendations: NoneNULL
```

```
Enter your choice: 6
```

```
Enter Source Product ID: 101
```

```
Enter Recommended Product ID: 110
```

```
⇒föü Recommendation Added!
```

```
Enter your choice: 7
```

```
PS C:\Users\koppu\OneDrive\Desktop\online_marker_place>
```

CONCLUSION

- **Simplifies buying and selling process**
- **Provides convenience and accessibility**
- **Supports digital economy growth**
- **Future scope:**
 - **AI-based recommendations**
 - **Faster delivery systems**
 - **Voice-based shopping**